

Statement of Work

Introduction

Field	Value
Name of Project	Messy APP
Sponsor	N/A
Stakeholder	Self-Directed Capstone
Project Manager	Dylan Sokolov
Team Member	Dylan Sokolov, Carlos Turcios, German Landaverde
Contractor/Vendor	N/A (Internal Senior Design Team)

Background

This project addresses the inefficiency of manual wardrobe management by introducing a digital inventory system. The solution leverages **RFID** technology to automate the tracking of clothing items. The app aims to provide users with a real-time, accurate overview of their closet contents and status (available/in use), reducing time spent organizing and deciding what to wear.

Purpose

The purpose of the SWO project is to design, develop, and test a functional **mobile application prototype** that serves as a proof-of-concept for an RFID-based wardrobe management system. The core goals are to validate the feasibility of using RFID for inventory tracking, establish a cloud-based data solution, and demonstrate an automated outfit generation feature.

Scope of Work

The core work involves building the mobile application prototype with NFC reading capabilities, a cloud database backend, and the Outfit Generator logic.

To Be Done (In-Scope Item)	Resources Needed	Outcome	Time Involved (Estimated)	General Steps

Mobile App Development	RFID tag ID validation and uniqueness enforcement	Functional app/website prototype for one platform (e.g., Android .apk)	Entire project duration (Phases 2-4)	UI/UX design, component development, state management setup.
NFC/RFID Integration	NFC-enabled mobile device, NFC tags/chips (min. 10), NFC SDK/API access	Seamless scanning to read unique item IDs and update item status.	Phase 2 (PoC), Phase 3 (Integration)	Implement reader logic, test tag association, link scan events to database updates.
Laundry Status	Laundry status toggle functionality	Valid toggle functionality	Phase 3	Toggle switch to mark items as “sent to laundry”
Inventory Database	PostgreSQL database	SQL database to store user input	Phase 1 (Schema), Phase 3 (Deployment)	Design schema, configure cloud hosting, set up CRUD operations.
Basic Outfit Generator	Application logic, Item data (color, type, status)	Feature suggesting complete, available outfits based on basic rules.	Phase 3	Define combination rules, implement selection algorithms, display suggestions.

Location

Site	Duration
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School	Periodic team meetings, technical demos.
Remote Location	Primary location for code development and individual work.

Schedule

Phases	Description	Start (Estimate)	Finish (Estimate)
Phase 1: Planning & Analysis	Define technical requirements, select technology stack, and create detailed design documents.	[Insert Start Date]	[End of Month 1]
Phase 2: Design & Prototyping	Design database schema, create UI, and establish NFC Proof-of-Concept.	[Start of Month 2]	[Mid-Month 2]
Phase 3: Development & Integration	Build core app modules, integrate NFC/DB, and develop Outfit Generator.	[Mid-Month 2]	[End of Month 3]
Phase 4: Testing & Deployment	Conduct unit testing, UAT, final bug fixes, and documentation preparation.	[Start of Month 4]	[Mid-Month 4]
Phase 5: Final Submission	Formal presentation and submission of all deliverables.	[Mid-Month 4]	[Insert Final Submission Date]

Deliverables

Deliverable	Objective	Due Date (Estimate)
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Functional Prototype Application	Working mobile website (v1.0) with all core features (NFC, Tracking, Generator)	[End of Phase 4]
Statement of Works	Complete, documented, and stable codebase on GitHub.	[End of Phase 4]
Final Project Documentation	Report detailing system architecture, testing, and future work.	[End of Phase 4]

Milestones

Milestones	Start Date (Estimate)	Finish Date (Estimate)
Tech Stack Selection Finalized	[Phase 1 Start]	[End of Phase 1]
NFC Read/Write Proof-of-Concept	[Phase 2 Start]	[Mid-Phase 2]
Full Inventory CRUD Functional	[Phase 3 Start]	[Mid-Phase 3]
Final Prototype Version 1.0 Complete	[Phase 4 Start]	[End of Phase 4]

Tasks (Initial Phase 1 & 2 Tasks)

Activity	Duration (Hours/Days)	Team Member
Finalize Tech Stack Decision	1 day	All Team Members
Database Schema Design	2 days	[Database Lead]
High-Fidelity UI/UX Mockups	4 days	[Design Lead]

NFC Communication Research & POC Setup	3 days	[Hardware Lead]
Write Final Project Report Draft (Sections 1-3)	5 days	All Team Members

Reporting and Communications

The team will hold **weekly sync-up meetings** to review progress against the schedule, identify roadblocks, and assign new tasks. A **weekly status report** will be submitted to the Project Sponsor outlining achievements, pending tasks, and any risks encountered. Communication will be primarily conducted via [Preferred Chat Platform, e.g., Discord/Slack/Email].

Standards and Testing

Standard/Test	Team Member	Due Date (Estimate)
Code Review Standard	All Team Members	Throughout Phases 3 & 4
NFC Functionality Test	[Hardware Lead]	End of Phase 2
User Acceptance Testing (UAT)	[Testing Lead] + 5 External Users	Phase 4
Data Integrity and Security Test	[Database Lead]	Phase 4

Define Success

The project is deemed successful when:

1. The application successfully demonstrates **100%** successful NFC scanning (demo) and association for a minimum sample set of **10 unique clothing items**.
2. The application can handle **100 inventory records** securely and stably.
3. The app achieves a user satisfaction rating of **4/5 or higher** based on external user testing.

Requirements

- **Equipment:** NFC-enabled Android or iOS smartphone (as the scanner), NFC tags/chips (minimum 10), and one dedicated computer for server-side hosting/database management.
- **Certifications:** N/A (Academic Project).
- **Skills:** Proficiency in [Selected Mobile Framework/Language], experience with NoSQL

database design (Firebase/Firestore), and familiarity with version control (Git).

Other

Out-of-Scope Confirmation: The project will *not* include commercial deployment, advanced AI/ML for image recognition, or integration with external weather APIs.

- User authentication and multi-user support
- RFID hardware integration (app accepts tag IDs, scanning handled externally)
- Offline functionality or native mobile apps
- AI-powered outfit recommendations
- Social sharing or wardrobe marketplace features
- Automated laundry service integrations
- Payment or subscription handling
- Data export/import functionality

Payments

Note: This section is typically used for commercial projects. For a senior design project, it covers budget/cost tracking.

Who	How Much	Payment Due
NFC Tags/Chips Procurement	\$2	Phase 1
Cloud Hosting (if applicable)	\$25	Phase 3
Total Project Budget	\$50	N/A

Closure

Document	Signed Off (Date)
Statement of Work (SOW)	11/30/25

Final Project Report	12/6/25
Code Review Sign-off	Dylan Sokolov